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$$z^3 = -1$$

$$-1 = \text{cis}(\pi)$$

$$\Rightarrow z_k = \text{cis}\left(\frac{\pi + 2k\pi}{3}\right), \quad k = 0, 1, 2$$

$$\Rightarrow z_0 = \text{cis}\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\Rightarrow z_1 = \text{cis}(\pi) = -1$$

$$\Rightarrow z_2 = \text{cis}\left(\frac{5\pi}{3}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow \boxed{z_0 = \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad z_1 = -1, \quad z_2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i}$$

References